

1	You are given N bank transactions. Each transaction contains: • Sender • Receiver • Amount • Timestamp A transaction is considered fraudulent if: 1. The Sender, Receiver, and Amount are the same as a previous transaction. 2. The difference between their timestamps is ≤ 60 seconds. Input: 5 ANU JON 200 1000 ANU JON 200 1050 RAM SAM 300 2000 ANU JON 200 2000 RAM SAM 300 2050 Output: ANU JON 200 1050 RAM SAM 300 2050												
2	A gym offers membership plans: <table> <thead> <tr> <th>Months</th> <th>Cost</th> </tr> </thead> <tbody> <tr> <td>12</td> <td>₹15000</td> </tr> <tr> <td>9</td> <td>₹12000</td> </tr> <tr> <td>6</td> <td>₹9000</td> </tr> <tr> <td>3</td> <td>₹5000</td> </tr> <tr> <td>1</td> <td>₹2000</td> </tr> </tbody> </table> Given N months, find the minimum cost to purchase membership. You can buy multiple plans. Input: 15 Output: 20000	Months	Cost	12	₹15000	9	₹12000	6	₹9000	3	₹5000	1	₹2000
Months	Cost												
12	₹15000												
9	₹12000												
6	₹9000												
3	₹5000												
1	₹2000												
3	A hot air balloon has a maximum weight capacity X kg. You are given the weights of N people. Find the maximum number of people that can travel in the balloon such that the total weight does not exceed X. To maximize the count, choose the lightest people first. Input: 6 (People) 4 2 7 1 3 5 10 (Value of X) Output: 4												

4	Given an array of integers (positive, negative, and zero), find the maximum product that can be obtained from any non-empty subset of the array. Input [-1, -1, -2, 4, 3] Output 24 Explanation: $(-1) \times (-1) \times 4 \times 3 = 12$ $(-1) \times (-2) \times 4 \times 3 = 24 \leftarrow \text{Maximum}$
5	A laptop needs a minimum charge X to work. Count how many laptops can work (i.e., charge $\geq$ X). X = minimum charge required for a laptop to work Array A [] containing charge percentages of laptops Input: 30 10 40 50 20 35 Output: 3
6	You are given an integer n representing a number of people initially in happy state In each iteration • 70% happy becomes Sad • 30% of happy remains happy • 50% sad remains Sad • 50% sad becomes Happy Input: 100 3 Output: Happy: 41.2, Sad: 58.8
7	Given an array of N integers (which may contain positive, negative, or zero values), find the maximum possible sum of any contiguous subarray. A subarray is a continuous part of the array. Input: 8 -2 -3 4 -1 -2 1 5 -3 Output: 7
8	Given an array of N integers, find: • The element with the minimum frequency. • The element with the maximum frequency. If multiple elements have the same frequency: • For maximum and minimum frequency, return the element that appears first in the array. Input: 8 1 2 2 3 3 3 4 4 Output: Minimum Frequency Element = 1 Maximum Frequency Element = 3
9	There are N people standing in a circle, numbered from 1 to N. Starting from person 1, every K-th person is eliminated from the circle. After a person is removed, counting continues from the next remaining person. This process continues until only one person remains.

	Your task is to find the number of the last remaining person. Input: 8 4 Output: 6
10	You are given a sorted array of integers of size N and a target value X. (Use Binary Search only) Your task is to find: 1. The index of the first occurrence of X in the array. 2. The index of the last occurrence of X in the array. If X is not present in the array, return -1. Input: 8 1 2 2 2 3 4 4 5 2 Output: 1 and 3
11	You are given a string consisting of lowercase English letters. Encode the string as follows: • For characters at even indices (0-based), shift the character forward by 2 positions in the alphabet (+2). • For characters at odd indices, shift the character backward by 1 position in the alphabet (-1). • Alphabet wrapping is allowed: ◦ 'y' + 2 = 'a' ◦ 'z' + 2 = 'b' ◦ 'a' - 1 = 'z' Return the encoded string. Input: abcdef Output: caecge
12	You are given two integers N and S. Starting from S, find the first prime number greater than or equal to S. Then repeatedly: • Take the next two prime numbers. • Find their sum. • Continue this process until the sum becomes greater than or equal to N. Print the final sum obtained. Input: N = 50 S = 10 Output: 52 Explanation: 11 + 13 = 24 17 + 19 = 36 23 + 29 = 52 > 50 Hence 52 is answer
13	You are given an array of size N containing numbers from 1 to N. One number is missing and one number is repeated exactly once.

	Find the repeating number and the missing number. Input: N = 5 arr = [1, 2, 2, 4, 5] Output: Repeating = 2, Missing = 3
14	Array A is given in ascending order Array B is given in descending order Create a new array of n (size of A) + m (Size of B) Containing all elements from both arrays in ascending order.
15	Alice and her friends are playing a game of verbal Kho-Kho. Alice is acting as a mediator, and the rest of the N friends are seated on N chairs, one each. Alice starts by providing a paper with a single-digit number to the friend present at number 1. Let's denote friends by F, where F will be of size N. F[1]...F[N] represents friends seated respectively. After receiving the paper with a digit, F[1] will enact and try to tell F[2] without speaking. Similarly, F[2] will communicate to the next person i.e., F[3]. This continues until the last person F[N] understands the digit. Finally, the last person will write the digit on a separate paper and give it to Alice to compare both papers. If the digits are similar then, Alice will give a T-shirt to each friend. However, if the digits do not match, Alice will ask each friend's digits, and she will offer the T-shirts to only those who understood the digits correctly. Given N number of friends and digit array D, denoting the digit understood by each friend F. finds out how many of Alice's friends have not enacted well OR did not understand the enactment by the previous friend correctly. Example 1: 3 -> N, number of friends 4 4 4 – array D. denoting digit understanding by N friends Output: 0 Explanation: All of them understood the digits correctly. Example 2: 5 1 2 3 2 2 Output: 4 Explanation: 1st, 2nd, 3rd, and 4th friends could not enact OR understand the enactment.

16	A group of explorers discovers several treasure chests inside an ancient cave. Each chest contains a certain number of gold coins. The chief explorer has a magical lock that can only be opened if the total number of coins in exactly two chests equals a target value K. Your task is to find the indices of the two chests whose coin values add up to K. It is guaranteed that exactly one valid pair exists. You may not use the same chest twice. Return the indices of the two chests in any order. Time Complexity: $O(N)$ Input: 4 2 7 11 15 9 Output: 0 1
17	Given an array of N integers, remove all duplicate elements while keeping only the first occurrence of each element. After removing the duplicates, reverse the resulting array and print it.
18	At the security checkpoint, airport security personnel have seized a number of travellers' belongings. Everything has been thrown into a big box (array). Each product carries a specific level of risk[0,1,2]. The risk severity of the items in this case is represented by an array[] of N integer values. Sorting the elements in the array according to the degrees of danger is the task at hand. Between 0 and 2 are the risk values. Example : Input : 7 -> Value of N [1,0,2,0,1,0,2]-> Element of arr[0] to arr[N-1], while input each element is separated by new line. Output : 0 0 0 1 1 2 2 -> Element after sorting based on risk severity Example 2: input: 10 -> Value of N [2,1,0,2,1,0,0,1,2,0] -> Element of arr[0] to arr[N-1], while input each element is separated by a new line. Output : 0 0 0 0 1 1 1 2 2 2 ->Elements after sorting based on risk severity. Explanation: In the above example, the input is an array of size N consisting of only 0's, 1's and 2s. The output is a sorted array from 0 to 2 based on risk severity.
19	Jack is a sports teacher at St. Patrick's School. He makes games not only to make the student fit but also smart. So, he lined up all the N number classes. of students in his class. At each position, he has fixed a board with the Integer number printed on it. Each number is unique and in exactly the range of N. Let us say there are 10 students, then the boards will be printed with numbers from 1 to 10 in a random order given

by the sequence $A[]$. As a rule, all students wear a jersey with their numbers printed on it. So if there are students, each will have a unique jersey number just like a football team. Now, in the beginning, all the students will stand as per the increasing order of their jersey numbers, from left to right. The only difference will be their respective board number which is placed at their respective location. The board location is fixed and cannot be changed. We can consider the arrangement as below. Suppose there are students, and the board is placed in the order of $[2\ 3\ 1\ 5\ 4]$.

Board — 2, 3, 1, 5, 4

Student's Jersey — 1, 2, 3, 4, 5

Now the game begins.

After every beat of the drum, each student will have to move to that location (index), where his board is pointing to. In the above case student with jersey #1 is standing with board #2, so now he will have to move to location #2. Similarly, all the other students will do.

So after the first beat of the drum, the alignment will be:

Board — 2, 3, 1, 5, 4.

This keeps going on and on until all the students are back to the way they were at the beginning. So, after 6 beats of the drum, all the students will be aligned the same way as before.

Given N and the order of the board of the respective positions, find the number of beats required to bring back the students to their original position.

So, for the above case, the answer is 6

Example 1:

Input:

3 -> Input integer, N

$\{1, 2, 3\}$ -> Input integer. $B[]$, board alignment.

Output:

1 -> Output

Explanation:

All the students will be standing in board positions;

Board — 1, 2, 3

Student's Jersey — 1, 2, 3

After the first beat of the drum:

Jersey #1 will move to index 1.

Jersey #2 will move to index 2.

Jersey #3 will move to index 3.

Hence, they will be back in their position in just 1 beat.

So, the answer is 1.

Example 2:

Input:

5 -> Input integer, N

{2, 3, 1, 5, 4} -> Input integer, B[], board alignment.

Output:

6 -> Output

Explanation:

All the students will be standing as below, with the board positions:

Board — 2, 3, 1, 5, 4

Student's Jersey — 1, 2, 3, 4, 5

After Beat-1 of the drum:

Jersey #1 has moved to index 2.

Jersey #2 has moved to index 3.

Jersey #3 has moved to index 1.

Jersey #4 has moved to index 5.

Jersey #5 has moved to index 4.

Board — 2, 3, 1, 5, 4

Student's Jersey — 3, 1, 2, 5, 4

After Beat-2 of the drum:

Jersey #3 has moved to index 2.

Jersey #1 has moved to index 3.

Jersey #2 has moved to index 1.

Jersey #5 has moved to index 5.

Jersey #4 has moved to index 4.

Board — 2, 3, 1, 5, 4

Student's Jersey — 2, 3, 1, 4, 5

After Beat-3 of the drum:

Board — 2, 3, 1, 5, 4

Student's Jersey — 1, 2, 3, 5, 4

After Beat-4 of the drum:

Board — 2, 3, 1, 5, 4

Student's Jersey — 3, 1, 2, 4, 5

After Beat-5 of the drum:

Board — 2, 3, 1, 5, 4

Student's Jersey — 2, 3, 1, 5, 4

After Beat-6 of the drum:

Board — 2, 3, 1, 5, 4

Student's Jersey — 1, 2, 3, 4, 5

Hence, they will be back in their positions after 6 beats.

	So, the answer is 6.
20	Bob is going to bet today on horse riding. There are N horses listed in a sequence of 1 to N. The probability of winning each horse is different so the prices for making a bet on the horses are not the same. There is no limit on the number of horses on which he can bet, but he thinks that if he bets on a continuous sequence of horses then he has a better chance to win. Bob will get K units of money if any horse on which he bets will win. But as the award is only K units so he wants to put money less than K. Bob wants to bet on as many horses as he can. As you are his best friend, he reached out to you for help, can you please find the length of the maximum continuous sequence of horses on which Bob can make a bet, and remember he will invest money less than K units. If there is more than one possible combination, Bob will bet randomly on any one of them. Given the number of horses(N), reward money(K), and price of betting on N horses in order. Hint: For each starting index of a horse, its end index in sequences will be equal to or greater than the end index of the previous starting index. Example 1: Input: 10 100 -> N = 10, K=100 30 40 50 20 20 10 90 10 10 10 -> Price to make bet on each horse in order Output: 3 Explanation: There are 10 horses, and the reward money is 100. So, Bob will put money in less than 100. There are two possible o sequences of length three whose total money for betting is less than 100, i.e. [50 20 20] (sum is 90) and [10 10 10] (sum is 30). Bob will choose randomly one sequence from these two. As none of the other sequences with a length greater than 3 will have a price less than 100 so the answer will be 3. Example 2: Input: 10 100 -> N = 10, K=100 10 90 80 20 90 60 40 60 70 75 -> Price to make bet on each horse in order Output: 1 Explanation: There are no two consecutive horses for which the sum of price is less than 100. So, Bob will choose randomly any one horse. And the max length of the sequence will be 1.

21	This vacation you went to visit the golden house. There are N rooms in this golden house and its owner needs someone to take care of the management of this house. As you have been unemployed for a long time, you are interested in this job. The owner of this house wanted an intelligent manager for this role, so he created one puzzle within that golden house. The person who will be able to solve that puzzle will be the manager of the golden house. The owner of the house kept some number of golden coins in each room. You have to choose two rooms, one from where you will enter and the other one from where you will exit. From any room either you can exit, or you can move to the next room. While visiting any room you will collect all the gold coins, and if you enter any room then you can't skip collecting gold coins from that room, you have to take those coins. The owner wants to have exactly K coins, when you exit the room, he guarantees that there will be at least one possible solution for this puzzle. Given several rooms (N), and several gold coins in N rooms. You have to find room numbers from where you will start and from where you will exit. If there is more than one solution possible, then you have to provide a solution with a smaller starting room number. You can assume that room numbers will start from 1 and end at N. Hint: Find a continuous subsequence whose sum will be exactly equal to K. Example 1: Input: 10 15 -> N =10, K = 15 5 3 7 14 18 1 18 4 8 3 -> Number of gold coins in each room. Output: 1 3 Explanation: There are ten rooms in the house. You want to have the Total sum of gold coins in a continuous sequence of rooms to be 15 There are two solutions to this i.e. [1, 3] and [8, 10] then the solution with a smaller starting room number will be printed hence [1, 3] is output.
22	For all of its products, a supermarket maintains a pricing structure. Each product has a value N printed on it. The price of the item is determined by multiplying the value N, which is read by the scanner, by the sum of all its digits. The goal here is to create software that, given the code of any item N, will compute the product (multiplication) of all the value digits (price). Example 1: Input : 5244 -> Value of N Output : 160 -> Price

	Explanation: From the input above Product of the digits 5,2,4,4 $5*2*4*4=160$ Hence, output is 160.
23	Jack is always excited about Sunday. It is favourite day, when he gets to play all day. And goes to cycling with his friends. So every time when the month starts he counts the number of Sundays he will get to enjoy. Considering the month can start with any day, be it Sunday, Monday.... Or so on. Count the number of Sunday Jack will get within n number of days. Example 1: Input mon-> input String denoting the start of the month. 13 -> input integer denoting the number of days from the start of the month. Output : 2 -> number of days within 13 days. Explanation: The month starts with mon(Monday). So the upcoming Sunday will arrive in next 6 days. And then next Sunday in next 7 days and so on. Now total number of days are 13. It means 6 days to first Sunday and then remaining 7 days will end up in another Sunday. Total 2 Sundays may fall within 13 days.
24	Given an integer array Arr of size N the task is to find the count of elements whose value is greater than all of its prior elements. Note : 1st element of the array should be considered in the count of the result. For example, Arr[]={7,4,8,2,9} As 7 is the first element, it will consider in the result. 8 and 9 are also the elements that are greater than all of its previous elements. Since total of 3 elements is present in the array that meets the condition. Hence the output = 3. Example 1: Input 5 -> Value of N, represents size of Arr 7-> Value of Arr[0] 4 -> Value of Arr[1] 8-> Value of Arr[2] 2-> Value of Arr[3] 9-> Value of Arr[4] Output : 3 Example 2: 5 -> Value of N, represents size of Arr 3 -> Value of Arr[0] 4 -> Value of Arr[1]

	5 -> Value of Arr[2] 8 -> Value of Arr[3] 9 -> Value of Arr[4] Output : 5 Constraints $1 \leq N \leq 20$ $1 \leq \text{Arr}[i] \leq 10000$
25	A furnishing company is manufacturing a new collection of curtains. The curtains are of two colors aqua(a) and black (b). The curtains color is represented as a string(str) consisting of a's and b's of length N. Then, they are packed (substring) into L number of curtains in each box. The box with the maximum number of 'aqua' (a) color curtains is labeled. The task here is to find the number of 'aqua' color curtains in the labeled box. Note: If 'L' is not a multiple of N, the remaining number of curtains should be considered as a substring too. In simple words, after dividing the curtains in sets of 'L', any curtains left will be another set(refer example 1) Example 1: Input: bbbaaababa -> Value of str 3 -> Value of L Output: 3 -> Maximum number of a's Explanation: From the input given above. Dividing the string into sets of 3 characters each Set 1: {b,b,b} Set 2: {a,a,a} Set 3: {b,a,b} Set 4: {a} -> leftover characters also as taken as another set Among all the sets, Set 2 has more number of a's. The number of a's in set 2 is 3. Hence, the output is 3. Example 2: Input : abbbaabbbb -> Value of str 5 -> Value of L

	Output: 2 -> Maximum number of a's Explanation: From the input given above, Dividing the string into sets of 5 characters each. Set 1: {a,b,b,b,b} Set 2: {a,a,b,b,b} Among both the sets, set 2 has more number of a's. The number of a's in set 2 is 2. Hence, the output is 2. Constraints: $1 \leq L \leq 10$ $1 \leq N \leq 50$ The input format for testing The candidate has to write the code to accept two inputs separated by a new line. First input- Accept string that contains character a and b only Second input- Accept value for N(Positive integer number) The output format for testing The output should be a positive integer number of print the message(if any) given in the problem statement.(Checkq the output in Example 1, Example 2).
26	An intelligence agency has received reports about some threats. The reports consist of numbers in a mysterious method. There is a number "N" and another number "R". Those numbers are studied thoroughly and it is concluded that all digits of the number 'N' are summed up and this action is performed 'R' number of times. The resultant is also a single digit that is yet to be deciphered. The task here is to find the single-digit sum of the given number 'N' by repeating the action 'R' number of times. If the value of 'R' is 0, print the output as '0'. Example 1: Input : 99 -> Value of N 3 -> Value of R Output : 9 -> Possible ways to fill the cistern. Explanation: Here, the number N=99 Sum of the digits N: $9+9 = 18$ Repeat step 2 'R' times i.e. 3 times $(9+9)+(9+9)+(9+9) = 18+18+18 = 54$

	Add digits of 54 as we need a single digit 5+4 Hence , the output is 9. Example 2: Input : 1234 -> Value of N 2 -> Value of R Output : 2 -> Possible ways to fill the cistern Explanation: Here, the number N=1234 Sum of the digits of N: 1+2+3+4 =10 Repeat step 2 'R' times i.e. 2 times (1+2+3+4)+(1+2+3+4)= 10+10=20 Add digits of 20 as we need a single digit. 2+0=2 Hence, the output is 2. Constraints: 0<N<=1000 0<=R<=50 The Input format for testing The candidate has to write the code to accept 2 input(s) First input- Accept value for N (positive integer number) Second input: Accept value for R(Positive integer number) The output format for testing The output should be a positive integer number or print the message (if any) given in the problem statement. (Check the output in Example 1, Example 2).
27	A chocolate factory is packing chocolates into the packets. The chocolate packets here represent an array of N number of integer values. The task is to find the empty packets(0) of chocolate and push it to the end of the conveyor belt(array). Example 1 : N=8 and arr = [4,5,0,1,9,0,5,0]. There are 3 empty packets in the given set. These 3 empty packets represented as 0 should be pushed towards the end of the array Input : 8 - Value of N [4,5,0,1,9,0,5,0] - Element of arr[0] to arr[N-1],While input each element is separated by newline

	Output: 4 5 1 9 5 0 0 0 Example 2: Input: 6 — Value of N. [6,0,1,8,0,2] – Element of arr[0] to arr[N-1], While input each element is separated by newline Output: 6 1 8 2 0 0
28	Joseph is learning digital logic subject which will be for his next semester. He usually tries to solve unit assignment problems before the lecture. Today he got one tricky question. The problem statement is “A positive integer has been given as an input. Convert decimal value to binary representation. Toggle all bits of it after the most significant bit including the most significant bit. Print the positive integer value after toggling all bits”. Constrains- 1<=N<=100 Example 1: Input : 10 -> Integer Output : 5 -> result- Integer Explanation: Binary representation of 10 is 1010. After toggling the bits(1010), will get 0101 which represents “5”. Hence output will print “5”.
29	An international round table conference will be held in india. Presidents from all over the world representing their respective countries will be attending the conference. The task is to find the possible number of ways(P) to make the N members sit around the circular table such that. The president and prime minister of India will always sit next to each other. Example 1: Input : 4 -> Value of N(No. of members) Output : 12 -> Possible ways of seating the members Explanation: 2 members should always be next to each other. So, 2 members can be in 2!ways Rest of the members can be arranged in (4-1)! ways.(1 is subtracted because the previously selected two members will be considered as single members now). So total possible ways 4 members can be seated around the circular table 2*6= 12. Hence, output is 12. Example 2:

	Input: 10 -> Value of N(No. of members) Output : 725760 -> Possible ways of seating the members Explanation: 2 members should always be next to each other. So, 2 members can be in 2! ways Rest of the members can be arranged in (10-1)! Ways. (1 is subtracted because the previously selected two members will be considered as a single member now). So, total possible ways 10 members can be seated around a round table is $2 \times 362880 = 725760$ ways. Hence, output is 725760. The input format for testing The candidate has to write the code to accept one input First input – Accept value of number of N(Positive integer number) The output format for testing The output should be a positive integer number or print the message(if any) given in the problem statement(Check the output in example 1, example2) Constraints : $2 \leq N \leq 50$
30	Particulate matters are the biggest contributors to Delhi pollution. The main reason behind the increase in the concentration of PMs include vehicle emission by applying Odd Even concept for all types of vehicles. The vehicles with the odd last digit in the registration number will be allowed on roads on odd dates and those with even last digit will on even dates. Given an integer array a[], contains the last digit of the registration number of N vehicles traveling on date D(a positive integer). The task is to calculate the total fine collected by the traffic police department from the vehicles violating the rules. Note : For violating the rule, vehicles would be fined as X Rs. Example 1: Input : 4 -> Value of N {5,2,3,7} -> a[], Elements a[0] to a[N-1], during input each element is separated by a new line 12 -> Value of D, i.e. date 200 -> Value of x i.e. fine Output : 600 -> total fine collected Explanation: Date D=12 means , only an even number of vehicles are allowed. Fine will be collected from 5,3 and 7 with an amount of 200 each. Hence, the output = 600. Example 2: Input : 5 -> Value of N

	{2,5,1,6,8} -> a[], elements a[0] to a[N-1], during input each element is separated by new line 3 -> Value of D i.e. date 300 -> Value of X i.e. fine Output : 900 -> total fine collected Explanation: Date D=3 means only odd number vehicles with are allowed. Find will be collected from 2,6 and 8 with an amount of 300 each. Hence, the output = 900 Constraints: • $0 < N \leq 100$ • $1 \leq a[i] \leq 9$ • $1 \leq D \leq 30$ • $100 \leq x \leq 5000$ The input format for testing The candidate has to write the code to accept 4 input(s). First input – Accept for N(Positive integer) values (a[]), where each value is separated by a new line. Third input – Accept value for D(Positive integer) Fourth input – Accept value for X(Positive integer) The output format for testing The output should be a positive integer number (Check the output in Example 1, Example e) if no fine is collected then print "0".
31	Given N gold wires, each wire has a length associated with it. At a time, only two adjacent small wires are assembled at the end of a large wire and the cost of forming is the sum of their length. Find the minimum cost when all wires are assembled to form a single wire. For Example: Suppose, Arr[]={7,6,8,6,1,1,} {7,6,8,6,1,1}-{7,6,8,6,2}, cost =2 {7,6,8,6,2}- {7,6,8,8}, cost = 8 {7,6,8,8} – {13,8,8}, cost=13 {13,8,8} -{13,16}, cost=16 {13, 16} – {29}, cost =29 2+8+13+16+29=68 Hence , the minimum cost to assemble all gold wires is 68. Constraints • $1 \leq N \leq 30$ • $1 \leq \text{Arr}[i] \leq 100$ Example 1: Input

	6 -> Value of N, represent size of Arr 7 -> Value of Arr[0], represent length of 1st wire 6 -> Value of Arr[1], represent length of 2nd wire 8 -> Value of Arr[2] , represent length of 3rd wire 6 -> Value of Arr[3], represent length of 4th wire 1 -> Value of Arr[4], represent length of 5th wire 1 -> Value of Arr[5], represent length of 6th wire Output : 68 Example 2: Input 4 -> Value of N, represents size of Arr 12 -> Value of Arr[0], represents length of 1st wire 2 -> Value of Arr[1], represent length of 2nd wire 2 -> Value of Arr[2], represent length of 3rd wire 5 -> Value of Arr[3], represent length of 4th wire Output : 34
32	
33	
34	